

Computer assignments 3 & 4
Mechanics of solids
-solid and structural mechanics

by

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Computer assignment 1

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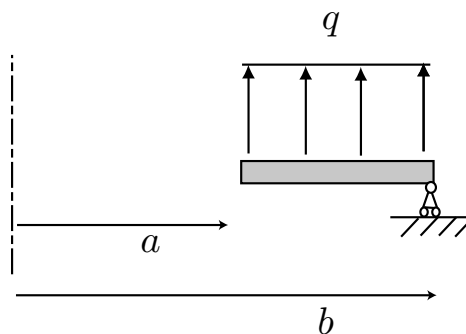
Computer assignment 2

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Computer assignment 3 An axisymmetric plate (see figure below) that is free at the inner radius and supported by a joint (allows no deformation but cannot resist any moment) at the outer radius is subjected to a vertical loading q . The plate has a thickness h , inner radius a , outer radius b , Young's modulus E , Poisson's ratio ν and yield stress σ_y . Compare the $\sigma_{rr}(r)$ for $z = h/2$ and the displacement $w(r)$ at $z = 0$ using

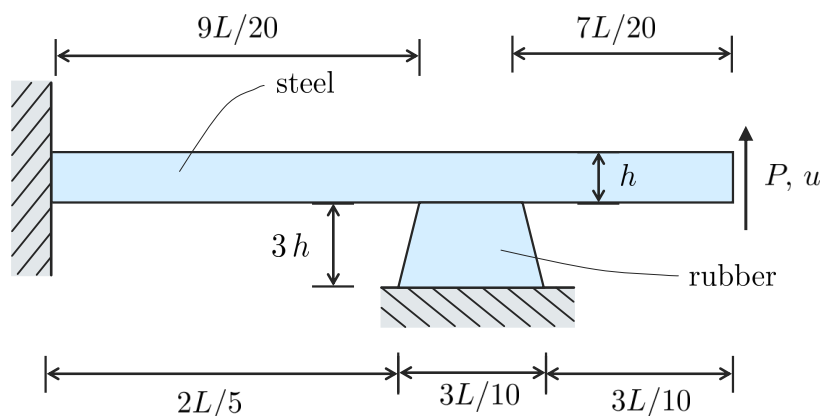
- (a) The axisymmetric plate model (use Matlab to solve for the numerical values of the integration constants and plot the result in Matlab).
- (b) Abaqus or your favorite FE program to solve the general equilibrium equation for axisymmetric conditions.

Investigate the behaviour and difference of the models. Here follows some suggestions of what you can analyse. By using model in (a): Investigate how much the maximum deflection and von Mises stress are changed if the thickness h would be decreased by a factor of 2. Are there any stress concentration in the FE model? What happens with the stress concentration when the mesh is refined (plot the von Mises stress by a color plot for two different meshes and compare the largest von Mises stress)?



Given data: $q = 10$ [MPa], $b = 0.2$ [m], $a = 0.1$ [m], $h = 0.02$ [m], $E = 210$ [GPa], $\nu = 0.3$ and $\sigma_y = 400$ [MPa]

Computer assignment 4 Assume that a steel beam and a rubber support are attached to each other (without relative moment) according to the figure below.



Both the steel beam and the rubber support have the thickness (into the plane) h . Plane stress can be assumed. The steel can be modelled as isotropic linear elastic while the rubber is modelled by the compressible neo-Hooke model.

Given: $L = 1$ m, $h = 30$ mm, $E_{\text{steel}} = 210 \cdot 10^3$ MPa, $\nu_{\text{steel}} = 0.3$, $E_{\text{rubber}} = 20$ MPa, $\nu_{\text{rubber}} = 0.45$,

- Simulate the assumed mechanical behaviour of the rubber material in Matlab/Python. For example by assuming incompressible behaviour and vary the stretch $0.1 \leq \lambda_1 \leq 1.9$ of a unidirectional loading (plot true stress vs stretch).
- Use Abaqus to determine how the deflection of the steel beam (vertical deflection of mid surface) varies along the beam for $P = 10$ kN. Is it a reasonable assumption with a small strain model in the steel beam and a large strain model in the rubber support (check this by contour plots of first and third principal strains)? Plot the deformed beam and rubber support.
- Use Abaqus to determine how the deflection of the steel beam (vertical deflection of mid surface) varies along the beam for $P = -10$ kN. Is there any difference compared to the result in (b)? Why?
- Compare results for the deflection in (b) and (c) with the solution from Euler-Bernoulli beam model without rubber support. Are the results above reasonable?